10-1-82

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, LAMP/RELAY, DIGITAL, KEY CODE ABF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic lamp or relay driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.7 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ABF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/152-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

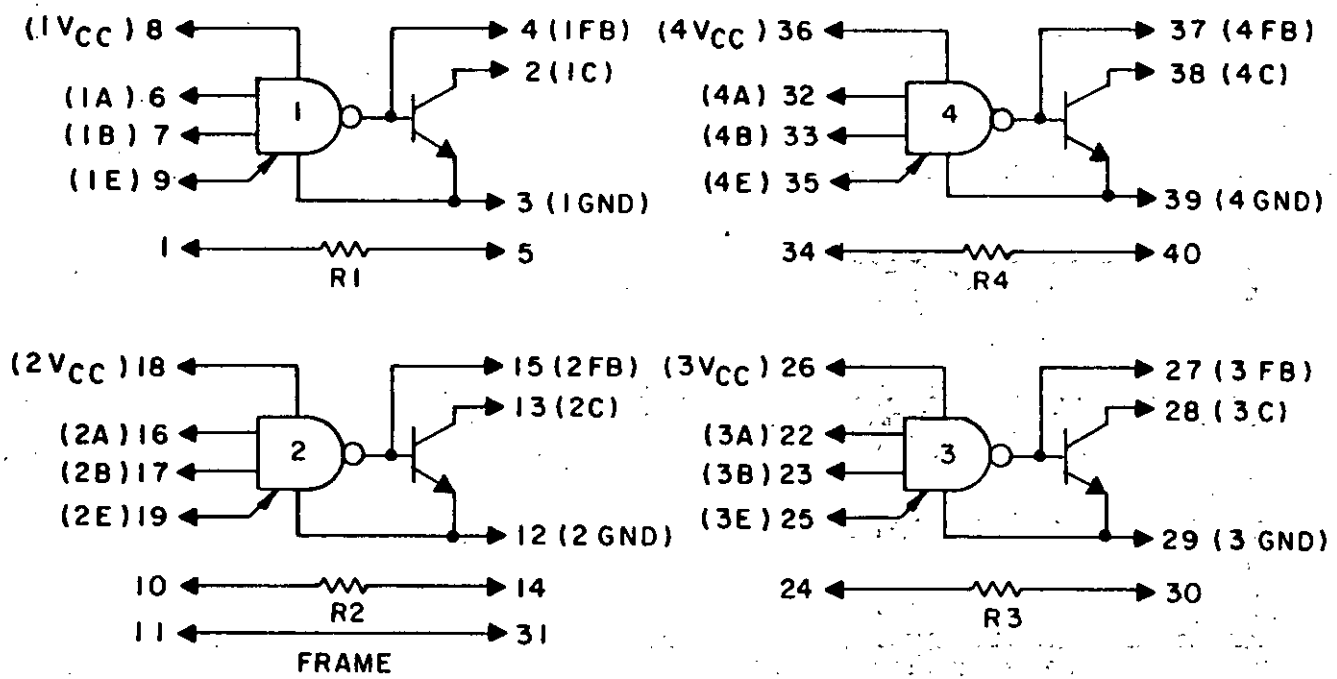
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. If test conditions A, B or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTES:
1. A = Gate input.
2. B = Gate input.
3. E = Expander input.
4. FB = Feedback node.
5. C = Open collector.
6. Pins 20 and 21 are not used.
7. R1 = R2 = 2.05 kΩ, 1/20 W.
8. R3 = R4 = 7.15 kΩ, 1/20 W.

FIGURE 1. ABF module function diagram, lamp/relay driver.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	Static output current (50 V)	II	S	0	1	0	5	0	5	0	5	µA
B1-B4	Collector to emitter current (50 V)	III	S	0	25	0	25	0	25	0	25	µA
C1-C8	Input current, low level (V_{CC} = 5.5 V)	IV	X	1.0	2.4	1.0	2.5	1.0	2.5	1.0	2.5	mA
C9-C12	Input current, low level (V_{CC} = 5.5 V)	IV	X	1.0	2.8	1.0	2.9	1.0	2.9	1.0	2.9	mA
C13-C20	Input current, low level (V_{CC} = 27.5 V)	IV	S	1.0	2.3	1.0	2.4	1.0	2.4	1.0	2.4	mA
C21-C24	Input current, low level (V_{CC} = 27.5 V)	IV	S	1.0	2.4	1.0	2.5	1.0	2.5	1.0	2.5	mA
D1-D8	Input current, high level	IV	X	0.0	0.1	0.0	0.5	0.00	0.75	0.00	0.75	µA
E1,E2	Resistance	V	X	2029	2070	1942	2158	2019	2081	1932	2168	Ω
E3,E4	Resistance	V	X	7078	7221	6775	7525	7043	7257	6739	7561	Ω
E5	Continuity	V	X	0	1	0	1	0	1	0	1	Ω
E6	Continuity	V	Q	0	1	0	1	0	1	0	1	Ω
F1-F16	Output voltage, low level (V_{CC} = 4.5 V)	VI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
F17-F28	Output voltage, low level (V_{CC} = 22.5 V)	VI	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
G1-G4	Output leakage current, high level	VII	X	0	25	0	25	0	25	0	25	µA
G5-G8	Output leakage current, high level	VII	S	0	25	0	25	0	25	0	25	µA
H1-H4	Power supply current, high level	IX	X	0	5	0	10	0	10	0	10	µA
H5-H8	Power supply current, high level	IX	S	0	25	0	50	0	50	0	50	µA
H9-H12	Power supply current, low level	IX	X	20	34	20	35	20	35	20	36	mA
H13-H16	Power supply current, low level	IX	S	20	28	20	28	20	28	20	28	mA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20 16:18Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1-J4	Transition time (t_{TLH})	X	S	0.10	1.30	0.10	1.40	0.10	1.40	0.10	1.40	µs
J5-J8	Transition time (t_{TLH})	X	S	0.10	1.10	0.10	1.20	0.10	1.20	0.10	1.30	µs
J9-J12	Transition time (t_{THL})	X	S	6	300	6	320	6	320	6	330	ns
J13-J16	Transition time (t_{THL})	X	S	6	200	6	210	6	210	6	220	ns
K1-K4	Propagation delay time (t_{PLH})	XI	S	0.50	4.20	0.50	4.40	0.50	4.40	0.50	4.50	µs
K5-K8	Propagation delay time (t_{PLH})	XI	S	0.50	7.50	0.50	7.90	0.50	7.90	0.50	8.00	µs
K9-K12	Propagation delay time (t_{PHL})	XI	S	10	500	10	525	10	525	10	550	ns
K13-K16	Propagation delay time (t_{PHL})	XI	S	10	400	10	425	10	425	10	440	ns
L1	Isolation, frame	XII	Q	10	∞	10	∞	10	∞	10	∞	MΩ
L2	Isolation, unused pins	XII	Q	10	∞	10	∞	10	∞	10	∞	MΩ
M1-M8	Input capacitance	XIII	Q	0	16	0	16	0	16	0	16	pF
N1-N4	Output crosstalk, high level	XIV	Q	0	550	0	550	0	550	0	550	mV
N5-N8	Output crosstalk, low level	XIV	Q	0	400	0	400	0	400	0	400	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through L, and N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 (condition A) and figure 3 (condition B). The power supply current shall be monitored during test and the reading shall not deviate more than ±10 percent from the initial current reading taken at the maximum class temperatures at the start of test.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through N shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/152 supersedes Department of the Navy, Naval Electronic Systems Command Drawing No. 28687-0102-871. Part number M28787/152-1 supersedes part numbers 0102-871 and M28787/152.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltages	5 Vdc ±10 percent or 25 Vdc ±10 percent (Possible by inserting an 825 ohm series dropping resistor per circuit)
Power dissipation per circuit (loaded) I_L = 260 mA, V_{IN} = 0 V	
V_{CC} = 5.5 V	0.33 W maximum
V_{CC} = 27.5 V	0.26 W maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

Power dissipation (output low level) I_L = 260 mA, V_{IN} = 0 V	
V_{CC} = 5.5 V	1.32 W maximum
V_{CC} = 27.5 V	0.84 W maximum
Power dissipation (output high level) V_{IN} open	
V_{CC} = 5.5 V	0.5 mW maximum
V_{CC} = 27.5 V	14 mW maximum
Input voltage (V_{CC} = 5 V ±10 percent)	Reference table VIII
Low level	0.80 V maximum
High level (V_{CC} = -1.5 V)	4.00 V minimum
Input voltage (V_{CC} = 25 V ±10 percent)	Reference table VIII
Low level	12.0 V maximum
High level (V_{CC} = -1.5 V)	26.0 V minimum
Output conditions	
Collector-emitter Breakdown voltage	50 V minimum
Transistor-ON current	260 mA maximum
Transistor-OFF current at 50 V_{CE}	25 µA maximum

6.5 Module function. The module consists of four independent and identical lamp or relay driver circuits and four pin programmable resistors.

6.5.1 Power conditions. Each drive circuit is designed to operate with 5 Vdc ±10 percent applied to the power pins (8, 18, 26, 36). Operation at 25 Vdc ±10 percent is possible by inserting an 825 ohm series dropping resistor between the 25 V power supply and the module power pin. A separate series dropping resistor is used for each individual driver circuit. NOTE: an 825 ohm resistor is available in the 10 ampere relay module (NRN). If the ABF and its associated driving circuit is operated at the same voltage (either 5 V or 25 V), then these circuits use a common supply. However, it is permissible to operate a lamp or relay driver at a lower voltage than its associated driving circuit in order to mechanize, for example, a positive-positive level shifter.

6.5.2 Input conditions. Each driver performs AND logic. A low level signal applied to either input turns on the output transistor; that is, any low level signal effects a switch closure (ON transistor) to ground at the output. A high level signal or an open circuit at all inputs effects an open switch (OFF transistor) at the output. An input expansion node is available in each circuit. This allows the mechanization of a multiple input circuit by utilizing external diodes.

6.5.3 Output conditions. Load power is supplied externally to appropriate module circuits (pins 2, 13, 28, and 38). The negative return of this power supply is connected directly to the appropriate circuit ground (pins 3, 12, 29, and 39). Each lamp or relay driver is capable of sinking any load less than 260 mA. If inductive loads (relays) are used, voltage spikes are suppressed to limit the V_{CE} voltage to less than 50 volts. If lamps are driven, bleeder resistors are employed to limit cold lamp inrush current to less than or equal to 600 mA. Output feedback nodes are accessible at pins 4, 15, 27, and 37. This allows the use of feedback capacitors between the output and feedback node in order to slow down the output stage. This may be desirable in mechanizing special interface circuits. Values of capacitance can be calculated only when exact requirements have been determined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

6.5.4 Programmable resistors. Each module provides two independently programmable 2.05 kΩ 1/20 W resistors. Program pins are 1, 5, 10, and 14. Each module provides two independently programmable 7.15 kΩ, 1/20 W resistors. Program pins are 24, 30, 34, and 40. When driving an ABF with a dedicated TTL gate, no pull-up resistor is required. In other applications where an additional load is connected to the TTL gate output, a module pull-up resistor may be necessary to insure a high level signal for the ABF.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N296)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

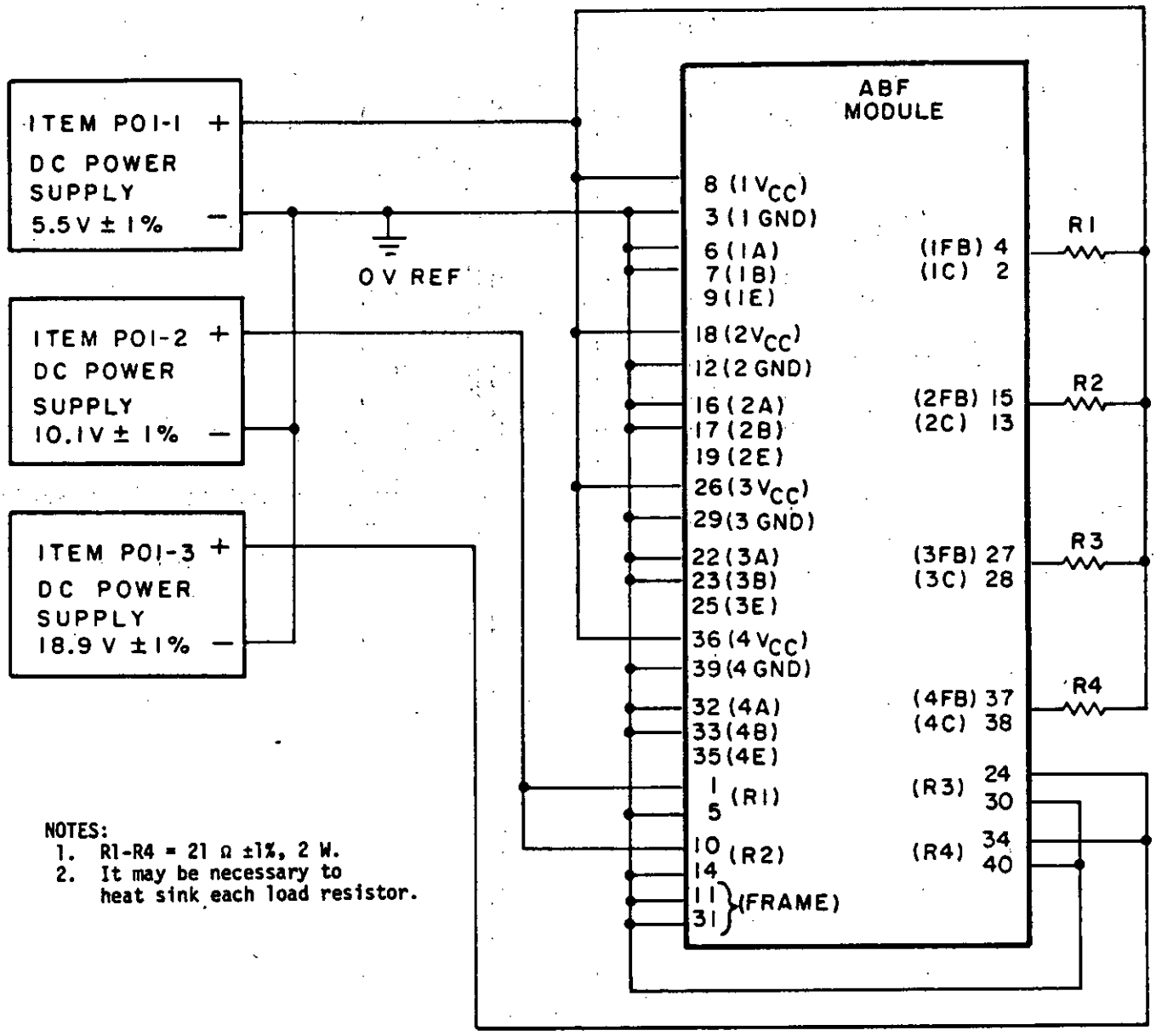
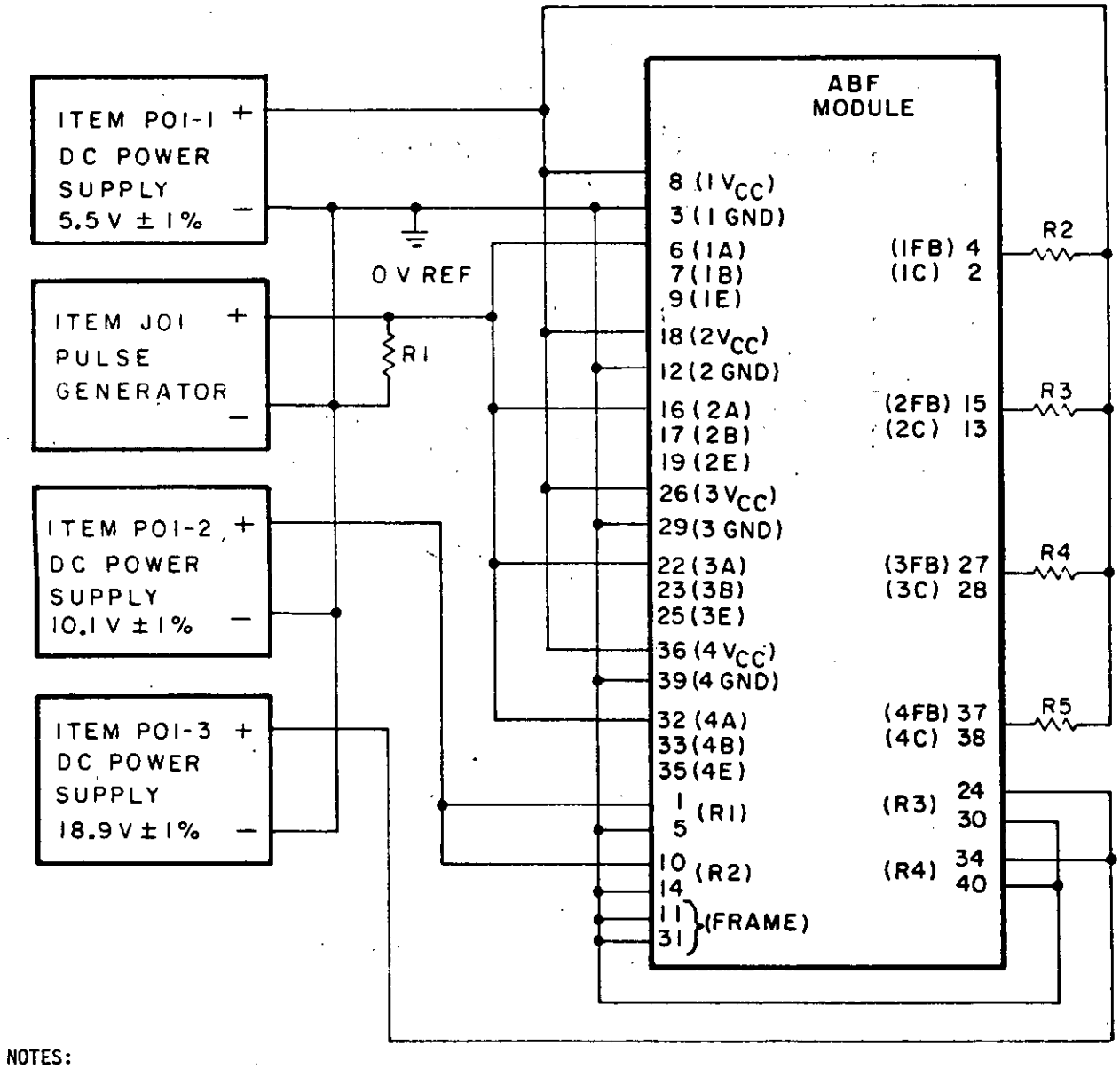


FIGURE 2. Operating temperature, component temperature and life test circuit (condition A).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50 Ω ±5%, 1 W.
2. R2-R5 = 8 Ω ±1%, 5 W.
3. Item J01 setup (10% duty cycle):

PW = 100 ±10 µs. F = 1.0 ±0.1 kHz.
$t_{TLH} = t_{THL}$ = 25 ±25 ns PA = 4.0 ±0.2 V.

FIGURE 3. Component temperature test circuit (test condition B).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

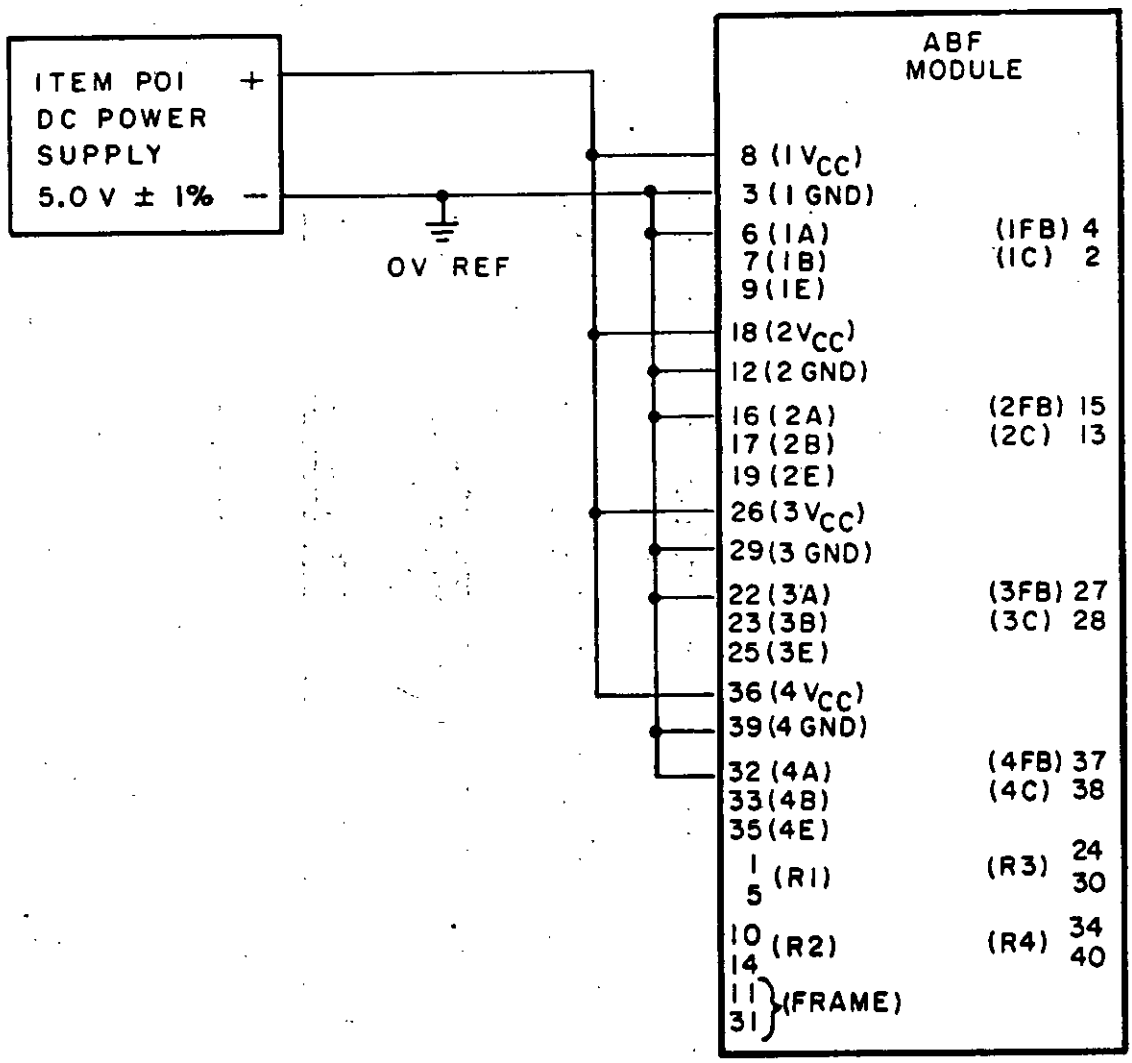


FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE II. Static input current (50V) test.

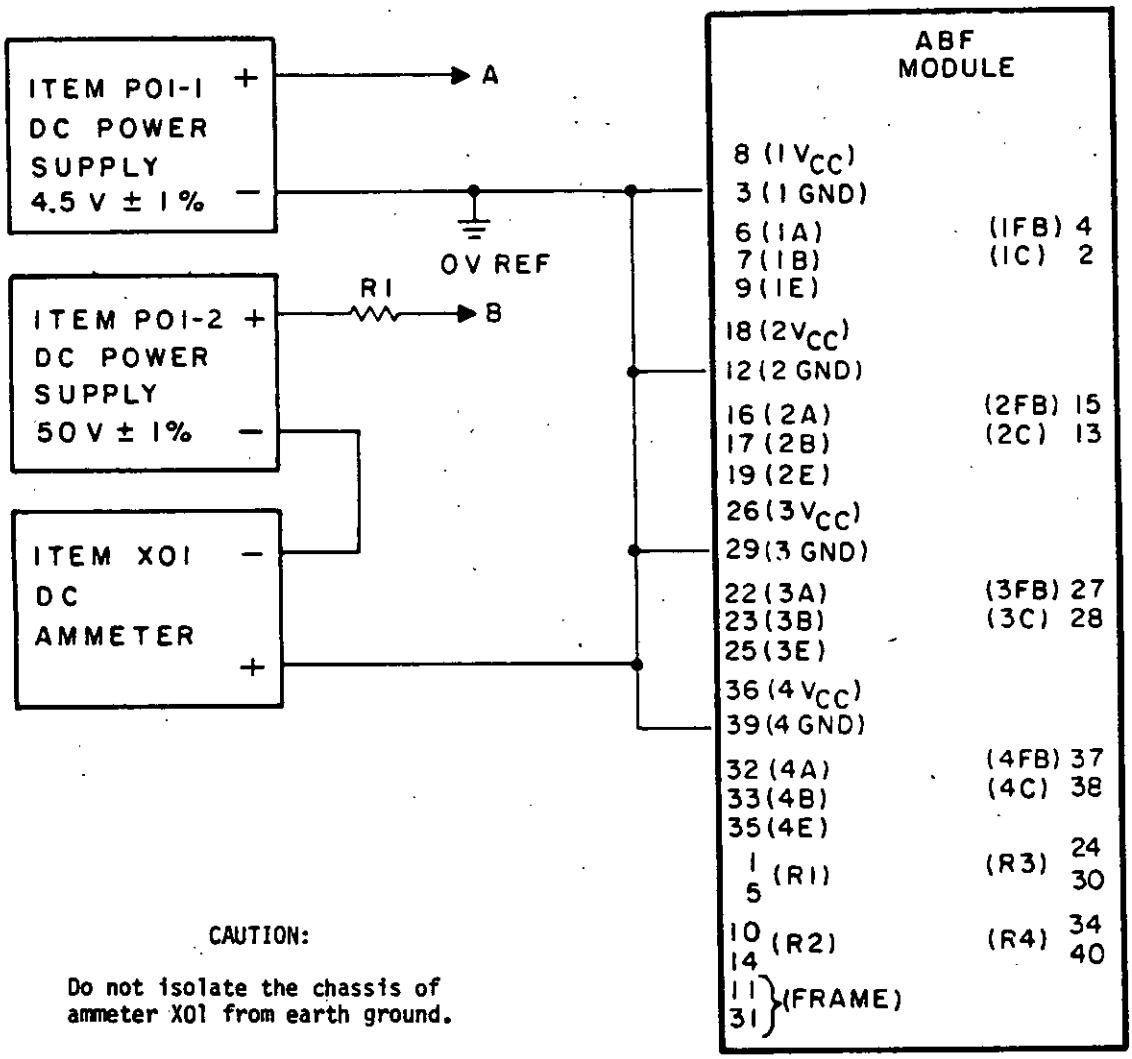
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	A1 Static input current (50V)	6	0	1	µA	5
A2	B1	7				
A3	A2	16				
A4	B2	17				
A5	A3	22				
A6	B3	23				
A7	A4	32				
A8	B4	33				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE III. Collector to emitter leakage current (50V) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	C1 Collector to emitter leakage current (50V)	8	2	0	25	µA	6
B2	C2	18	13				
B3	C3	26	28				
B4	C4	36	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



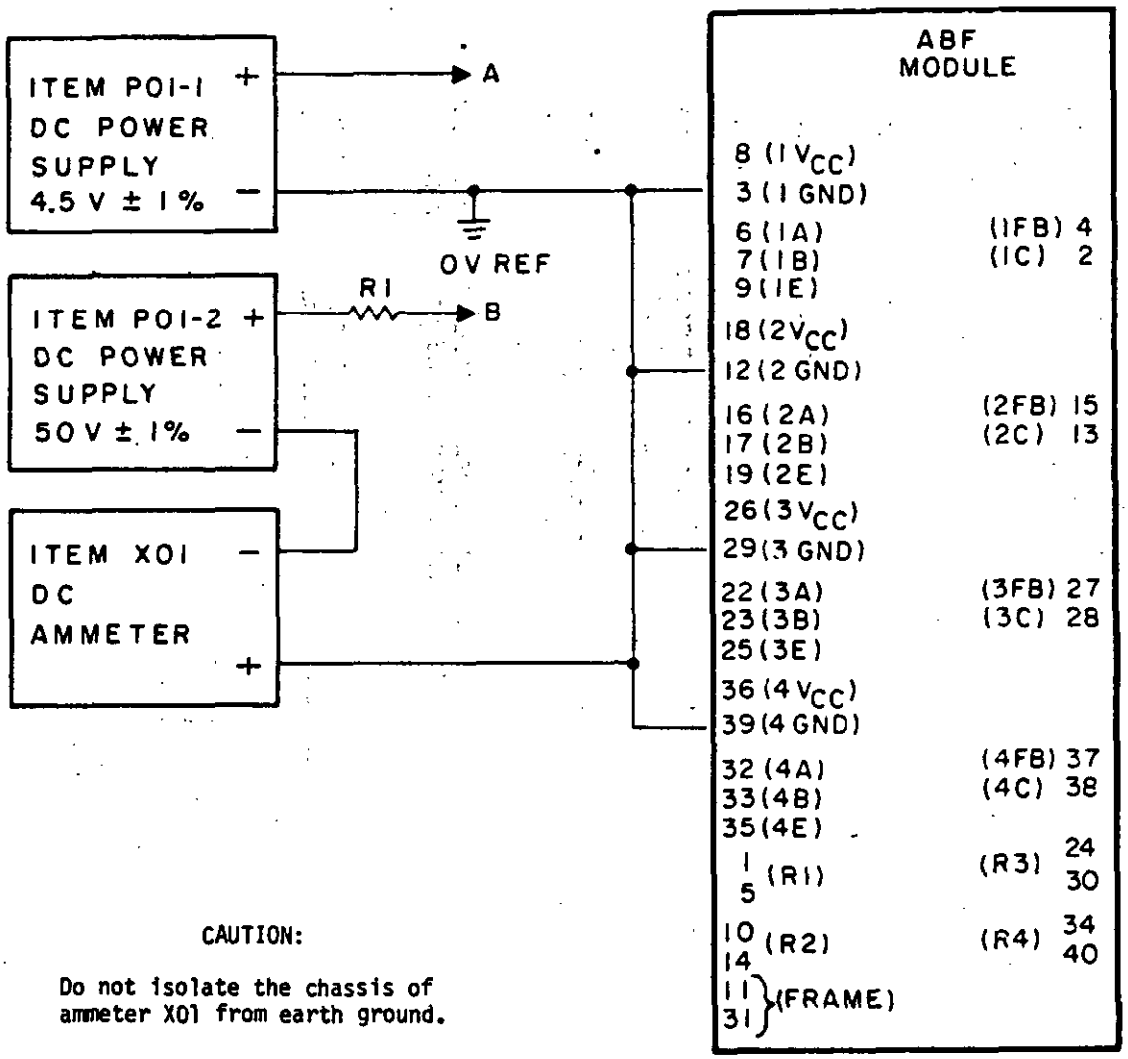
CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 5 kΩ ±1%, 1/8 W.

FIGURE 6. Collector to emitter leakage current (50 V) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 5 kΩ ±1%, 1/8 W.

FIGURE 6. Collector to emitter leakage current (50 V) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE IV. Input current (low-level and high-level) test.

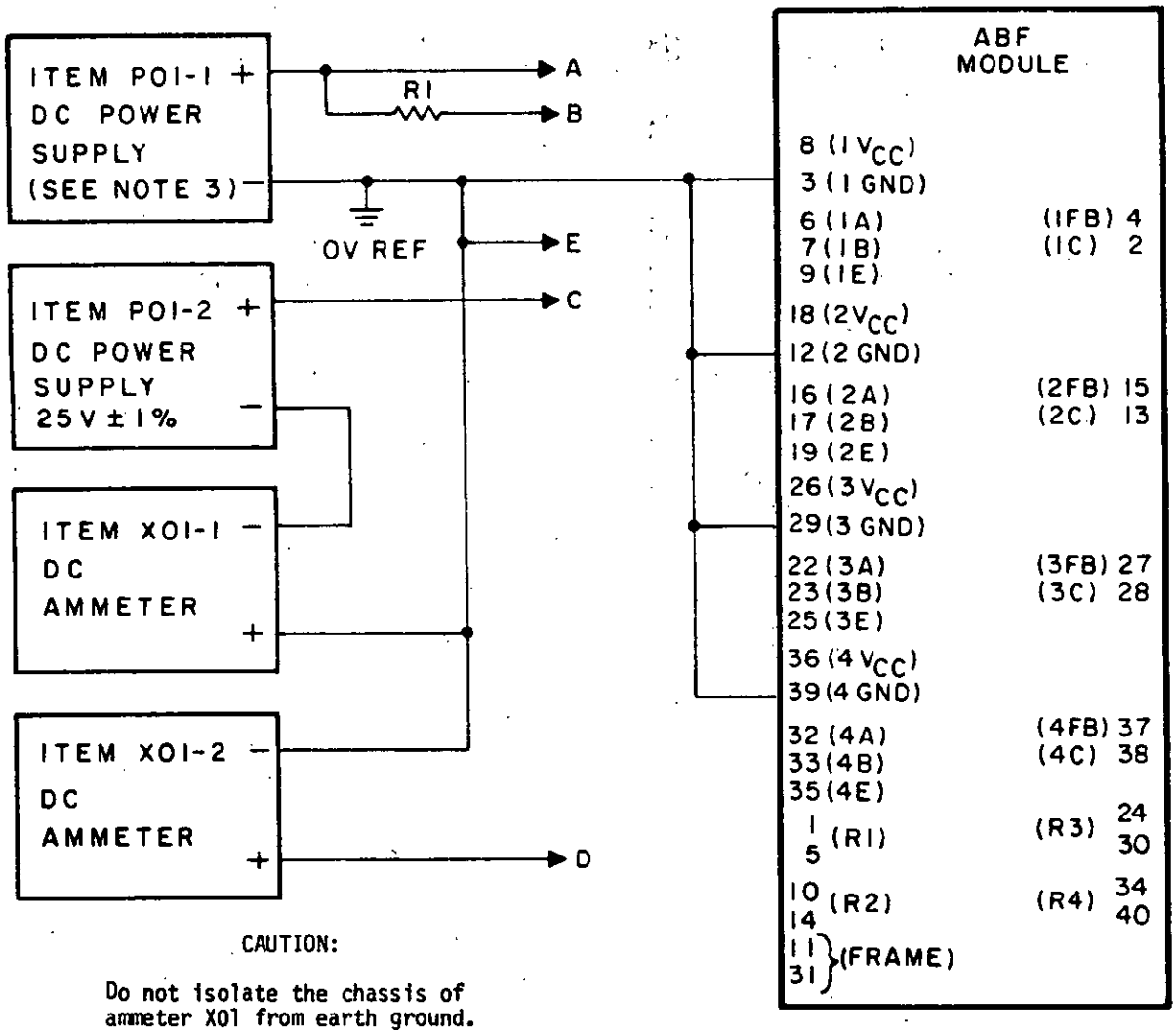
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
C1	A1 Input current, low level (V_{CC} = 5.5V)	8	Open	7	6	Open	1.0	2.4	mA	7
C2	B1	8		6	7					
C3	A2	18		17	16					
C4	B2	18		16	17					
C5	A3	26		23	22					
C6	B3	26		22	23					
C7	A4	36		33	32					
C8	B4	36		32	33					
C9	E1	8		6,7	9			2.8		
C10	E2	18		16,17	19					
C11	E3	26		22,23	25					
C12	E4	36		32,33	35					
C13	A1 Input current, low level (V_{CC} = 27.5V)	Open	8	7	6			2.3		
C14	B1		8	6	7					
C15	A2		18	17	16					
C16	B2		18	16	17					
C17	A3		26	23	22					
C18	B3		26	22	23					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE IV. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
C19	A4 Input current, low level (V_{CC} = 27.5V)	Open	36	33	32	Open	1.0	2.3	mA	7
C20	B4		36	32	33					
C21	E1		8	6,7	9			2.4		
C22	E2		18	16,17	19					
C23	E3		26	22,23	25					
C24	E4		36	32,33	35					
D1	A1 Input current, high level	8	Open	6	Open	7	0.0	0.1	µA	
D2	B1	8		7		6				
D3	A2	18		16		17				
D4	B2	18		17		16				
D5	A3	26		22		23				
D6	B3	26		23		22				
D7	A4	36		32		33				
D8	B4	36		33		32				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 825 Ω ±1%, 1/2 W.
2. Since items XO1-1 and XO2-2 are not used simultaneously, only one item XO1 is required.
3. Set item PO1-1 at 5.5 V ±1% for test conditions C1-C12 and D1-D8 and 27.5 V ±1% for test conditions C13-C24.

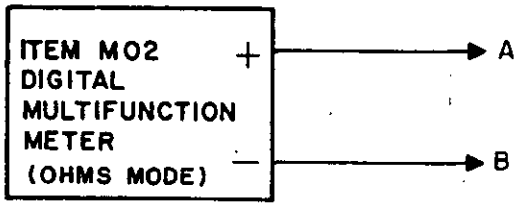
FIGURE 7. Input current (low-level and high-level) test circuit for table IV.

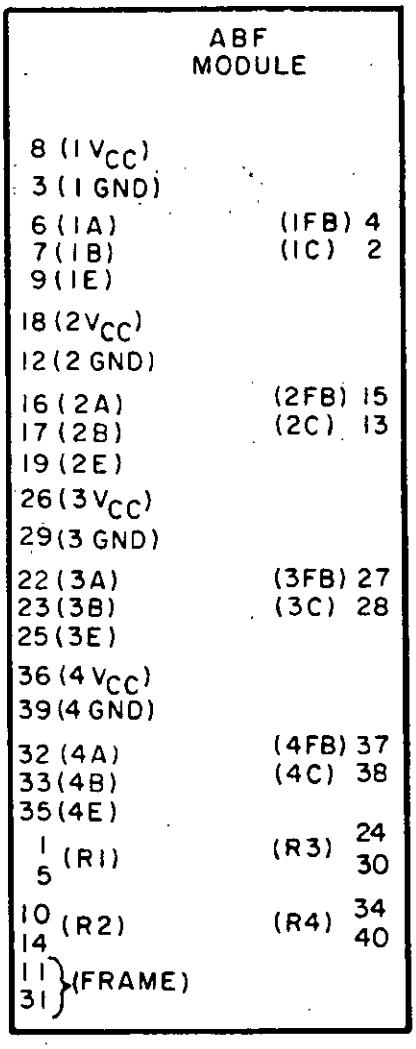
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE V. Resistance and continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	R1 Resistance	5	1	2029	2070	Ω	8
E2	R2	14	10	2029	2070		
E3	R3	30	24	7078	7221		
E4	R4	40	34	7078	7221		
E5	Continuity	31	11	0	1		
E6	Continuity	11,31	Frame	0	1		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.





NOTE: For test condition E6 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

FIGURE 8. Resistance and continuity test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (input threshold low-level) test.

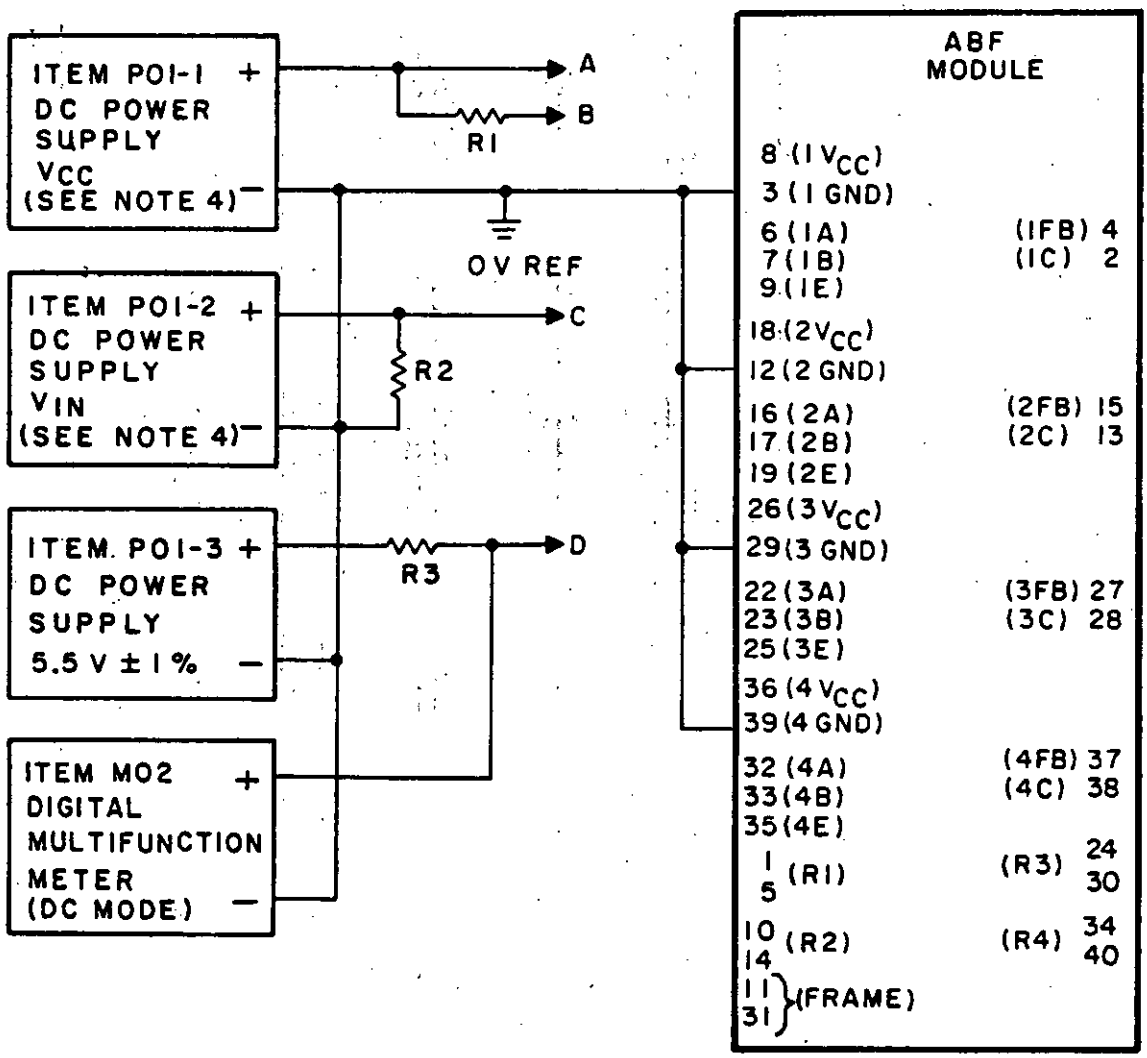
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1C Output voltage (A input threshold, low level; B input = 4.5 V)	7,8	Open	6	2	0.0	0.5	V	9
F2	2C	17,18		16	13				
F3	3C	23,26		22	28				
F4	4C	33,36		32	38				
F5	1C Output voltage (B input threshold, low level; A input = 4.5 V)	6,8		7	2				
F6	2C	16,18		17	13				
F7	3C	22,26		23	28				
F8	4C	32,36		33	38				
F9	1C Output voltage (A and B input thresholds, low level)	8		6,7	2				
F10	2C	18		16,17	13				
F11	3C	26		22,23	28				
F12	4C	36		32,33	38				
F13	1C Output voltage (E input threshold, low level; A and B inputs = 4.5 V)	6,7,8		9	2				
F14	2C	16,17,18		19	13				
F15	3C	22,23,26		25	28				
F16	4C	32,33,36		35	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (input threshold low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F17	1C Output voltage (A input threshold, low level; B input = 22.5 V)	Open	7,8	6	2	0.0	0.5	V	9
F18	2C		17,18	16	13				
F19	3C		23,26	22	28				
F20	4C		33,36	32	38				
F21	1C Output voltage (B input threshold, low level; A input = 22.5 V)		6,8	7	2				
F22	2C		16,18	17	13				
F23	3C		22,26	23	28				
F24	4C		32,36	33	38				
F25	1C Output voltage (E input threshold, low level; A and B inputs = 22.5 V)		6,7,8	9	2				
F26	2C		16,17,18	19	13				
F27	3C		22,23,26	25	28				
F28	4C		32,33,36	35	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20 16:18Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 825 Ω ±1%, 1/2 W.
2. R2 = 50 Ω ±5%, 1 W.
3. R3 = 21 Ω ±1%, 2 W.
4. Adjust V_{CC} and V_{IN} as shown in table VIII.

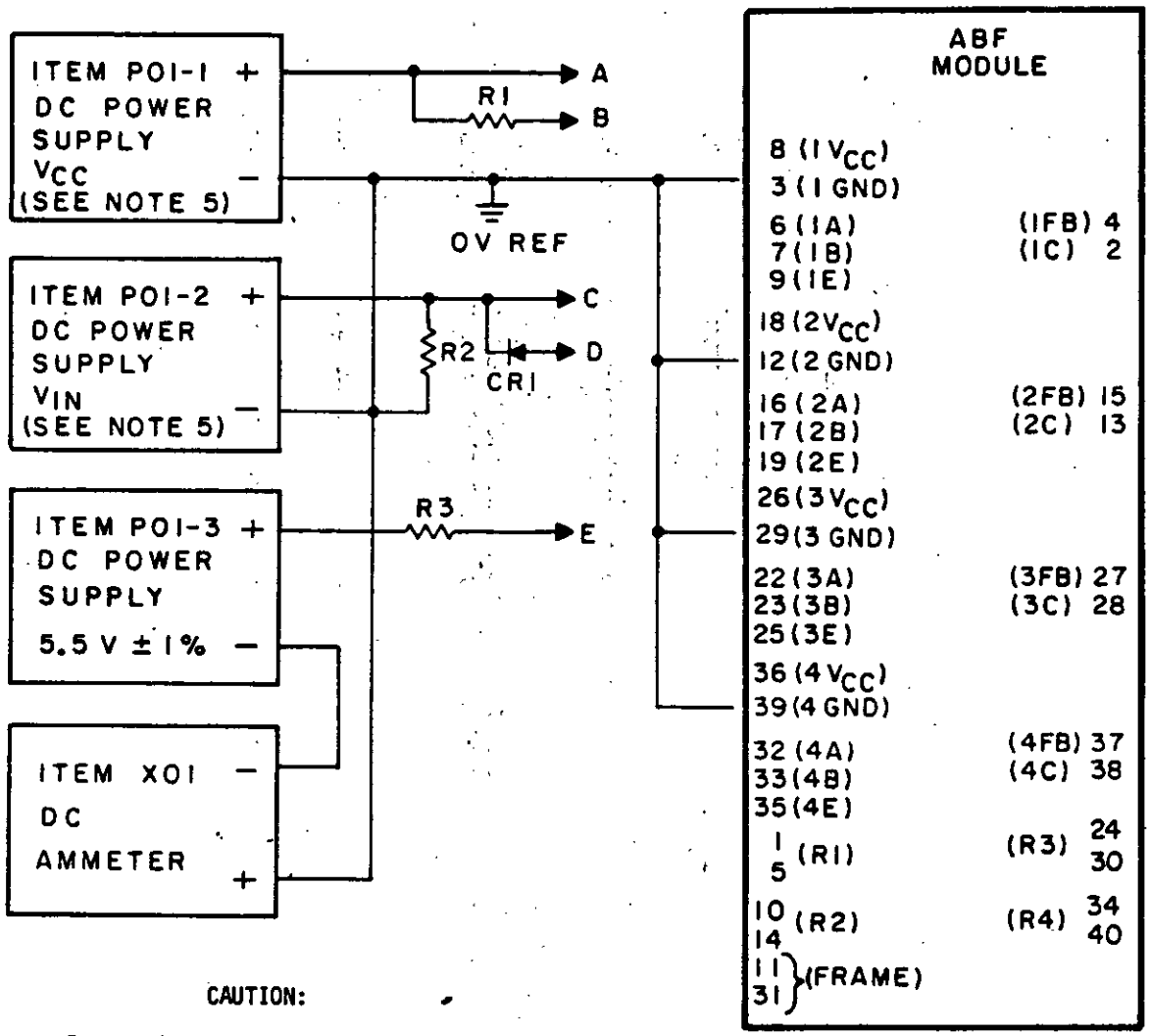
FIGURE 9. Output voltage (input threshold low-level) test for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE VII. Output leakage current (input threshold high-level) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	1C Output leakage current (A, B and E input threshold, high level)	8	Open	6,7	9	2	0	25	µA	10
G2	2C	18		16,17	19	13				
G3	3C	26		22,23	25	28				
G4	4C	36		32,33	35	38				
G5	1C	Open	8	6,7	9	2				
G6	2C		18	16,17	19	13				
Q7	3C		26	22,23	25	28				
G8	4C		36	32,33	35	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 825 Ω ±1%, 1/2 W.
2. R2 = 50 Ω ±5%, 1 W.
3. R3 = 21 Ω ±1%, 2 W.
4. CR1 = 1N4150 or equivalent.
5. Adjust V_{CC} and V_{IN} as shown in table VIII.

FIGURE 10. Output leakage current (input threshold high-level) test for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input threshold voltages (low-level and high-level) test.

Test condition	ABE input	Logic level	V_{CC} voltage (±5%) 1/	Acceptance requirements 25° ±5°C ambient temperature 1/		0° to 60°C 1/ Operating temperature	
				Initial	End-of-Life	Initial	End-of-Life
F1-F12	A and B	Low	4.5 V	1.30	1.10	1.00	0.80
F13-F16	E	Low	4.5 V	1.90	1.70	1.60	1.50
F17-F24	A and B	Low	22.5 V	14.5	13.3	13.3	12.0
F25-F28	E	Low	22.5 V	15.0	13.8	13.8	12.8
G1-G4	ALL	High	5.5 V	3.50	3.70	3.9	4.00
G5-G8	ALL	High	27.5 V	25.5	25.7	25.9	26.0
		High	V_{CC}	V_{CC} -2.0 V	V_{CC} -1.8 V	V_{CC} -1.6 V	V_{CC} -1.5 V

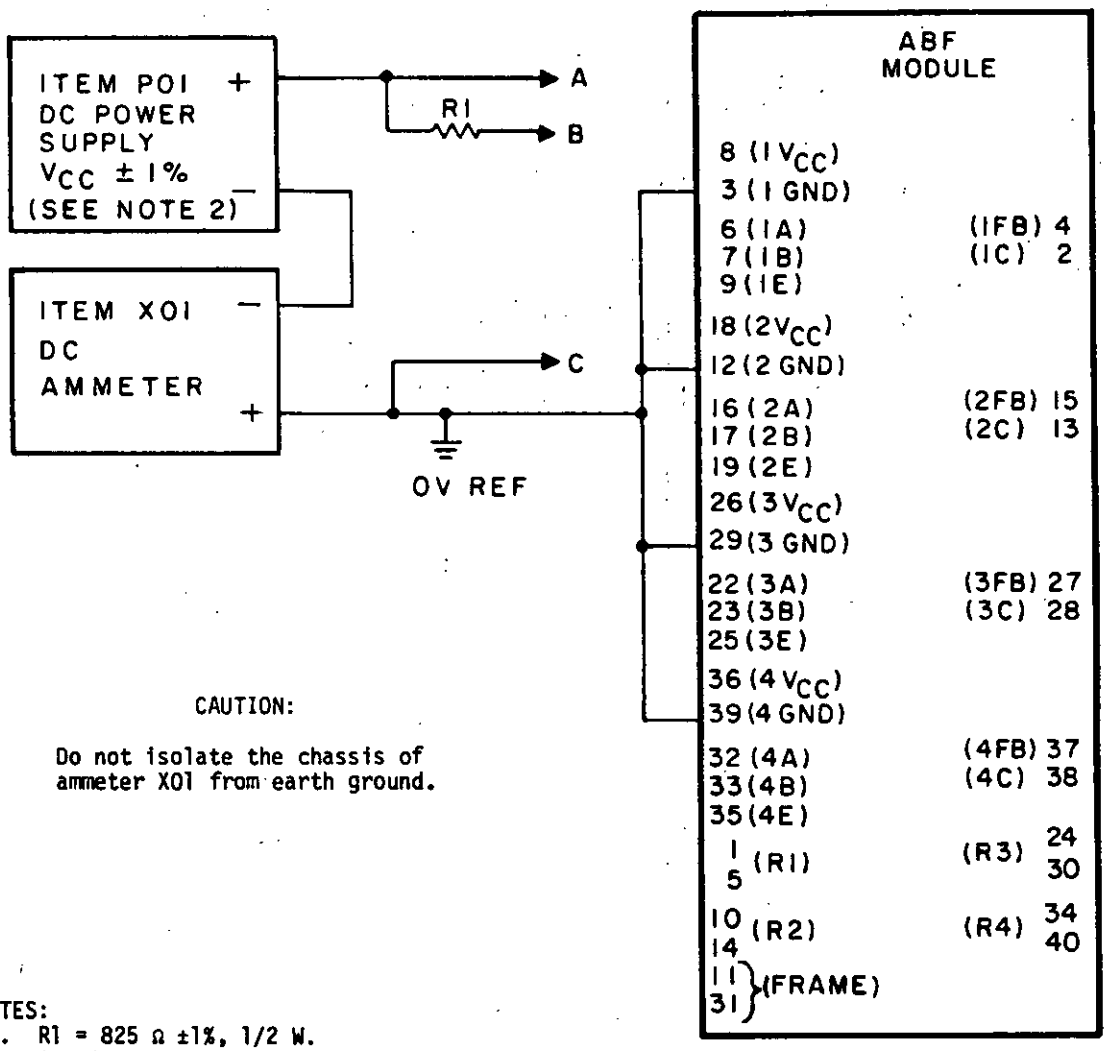
1/ When performing tables VI, VII and figures 9 and 10 tests, adjust item P01-1 voltage (V_{CC}) as specified in the V_{CC} column and adjust item P01-2 voltage (V_{IN}) as specified in this table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20 16:18Z
Check the source to verify that this is the current version before use.

TABLE IX. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Power supply current, high level (circuit inputs open)	8	Open	Open	0	5	µA	11
H2		18						
H3		26						
H4		36						
H5		Open	8			25		
H6			18					
H7			26					
H8			36					
H9	Power supply current, low level (circuit inputs GND)	8	Open	6	20	34	mA	
H10		18		16				
H11		26		22				
H12		36		32				
H13		Open	8	6		28		
H14			18	16				
H15			26	22				
H16			36	32				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 825 Ω ±1%, 1/2 W.
2. Set item P01 at V_{CC} = 5.5 V ±1% for test conditions H1-H4 and H9-H12 and V_{CC} = 27.5 V ±1% for test conditions H5-H8 and H13-H16.

FIGURE 11. Power supply current (high-level and low-level) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) test.

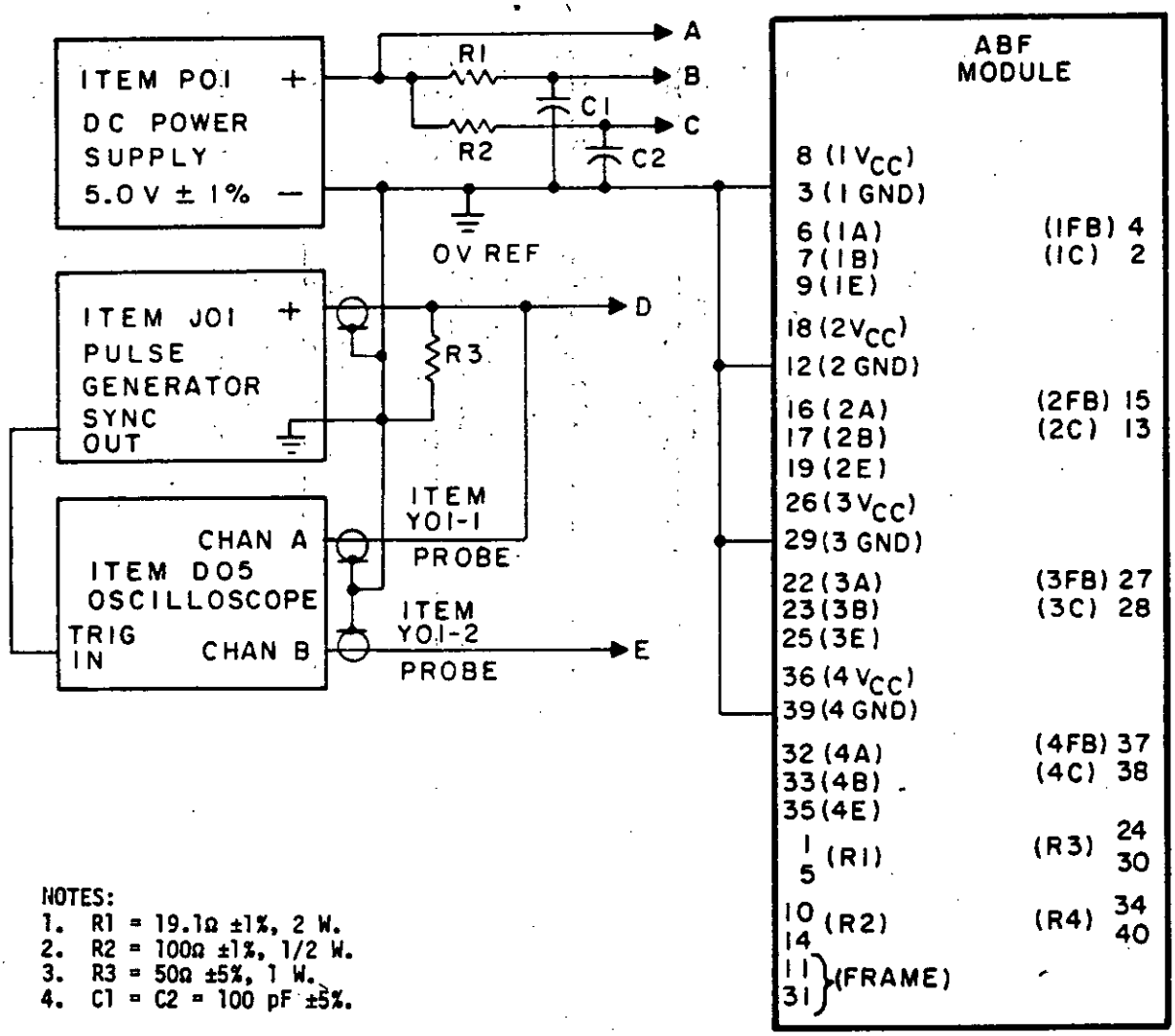
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
J1	1C Transition time (t_{TLH})	8	2	Open	6,7	2	0.10	1.30	µs	12,13
J2	2C	18	13		16,17	13				
J3	3C	26	28		22,23	28				
J4	4C	36	38		32,33	38				
J5	1C	8	Open	2	6,7	2		1.10		
J6	2C	18		13	16,17	13				
J7	3C	26		28	22,23	28				
J8	4C	36		38	32,33	38				
J9	1C Transition time (t_{THL})	8	2	Open	6,7	2	6	300	ns	
J10	2C	18	13		16,17	13				
J11	3C	26	28		22,23	28				
J12	4C	36	38		32,33	38				
J13	1C	8	Open	2	6,7	2		200		
J14	2C	18		13	16,17	13				
J15	3C	26		28	22,23	28				
J16	4C	36		38	32,33	38				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE XI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	1C Propagation delay time (t_{PLH})	8	2	Open	6,7	2	0.50	4.20	µs	12,13
K2	2C	18	13		16,17	13				
K3	3C	26	28		22,23	28				
K4	4C	36	38		32,33	38				
K5	1C	8	Open	2	6,7	2		7.50		
K6	2C	18		13	16,17	13				
K7	3C	26		28	22,23	28				
K8	4C	36		38	32,33	38				
K9	1C Propagation delay time (t_{PHL})	8	2	Open	6,7	2	10	500	ns	
K10	2C	18	13		16,17	13				
K11	3C	26	28		22,23	28				
K12	4C	36	38		32,33	38				
K13	1C	8	Open	2	6,7	2		400		
K14	2C	18		13	16,17	13				
K15	3C	26		28	22,23	28				
K16	4C	36		38	32,33	38				

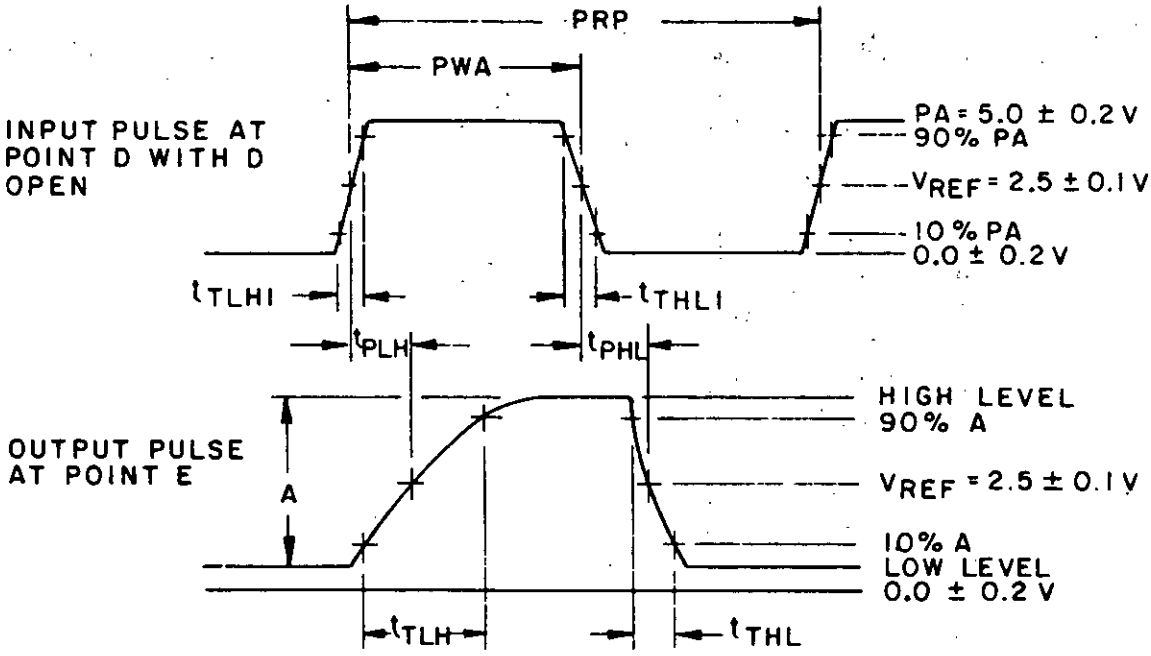
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 19.1Ω ±1%, 2 W.
2. R2 = 100Ω ±1%, 1/2 W.
3. R3 = 50Ω ±5%, 1 W.
4. C1 = C2 = 100 pF ±5%.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:

PRP = 30 ±1 µs.
PWA = 15 ±0.5 µs.
$T_{TLHI} = t_{THLI}$ = 25 ±5 ns.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

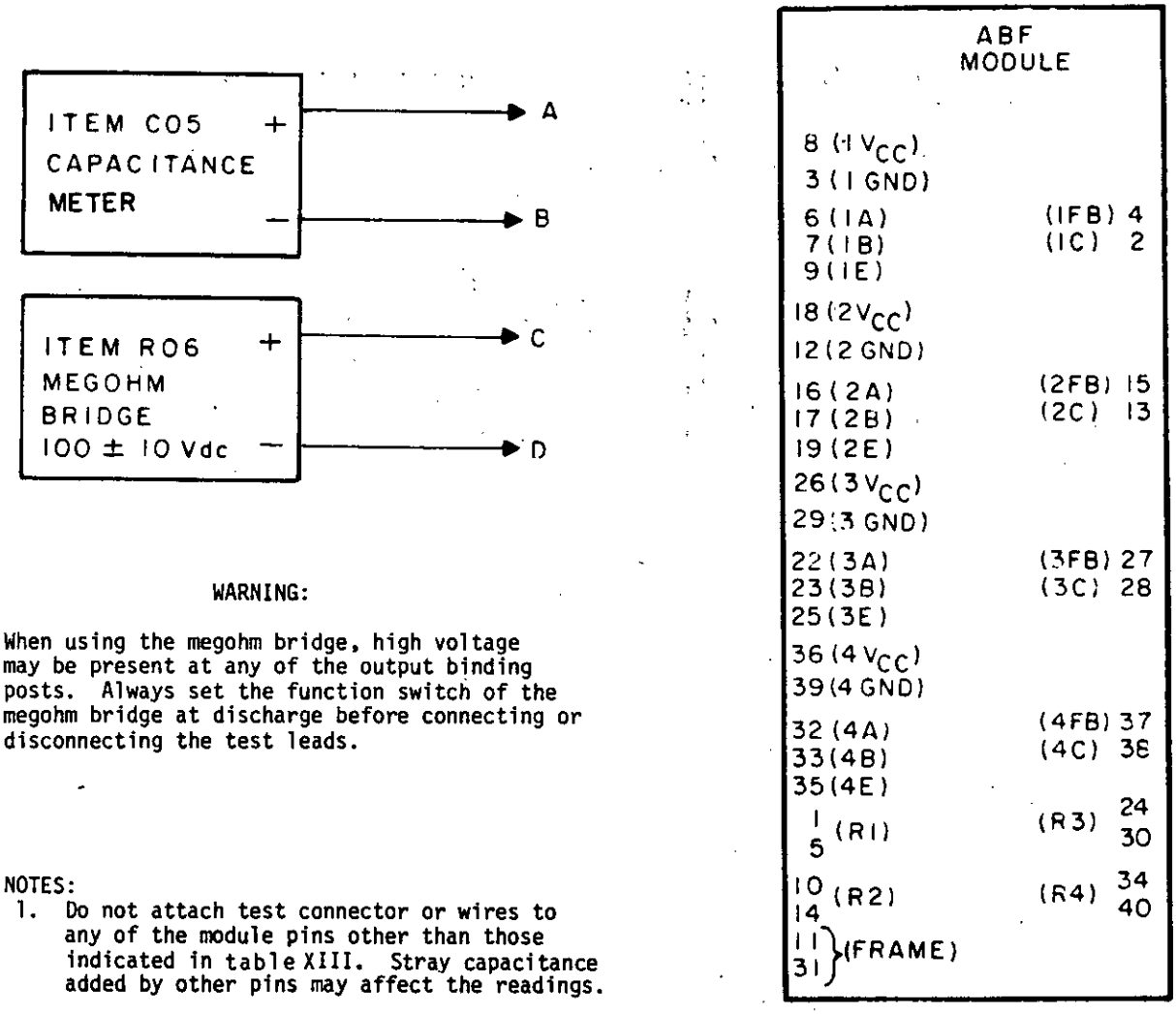
TABLE XII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
L1	Isolation, frame	All pins except 11 and 31	11, 31	10	∞	MΩ	14
L2	Isolation, unused pins	All pins except 20 and 21	20, 21	10	∞	MΩ	14

TABLE XIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1A Input capacitance	6	3	0	16	pF	14
M2	1B	7	3	↓	↓	↓	↓
M3	2A	16	12				
M4	2B	17	12				
M5	3A	22	29				
M6	3B	23	29				
M7	4A	32	39				
M8	4B	33	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at discharge before connecting or disconnecting the test leads.

NOTES:

1. Do not attach test connector or wires to any of the module pins other than those indicated in table XIII. Stray capacitance added by other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

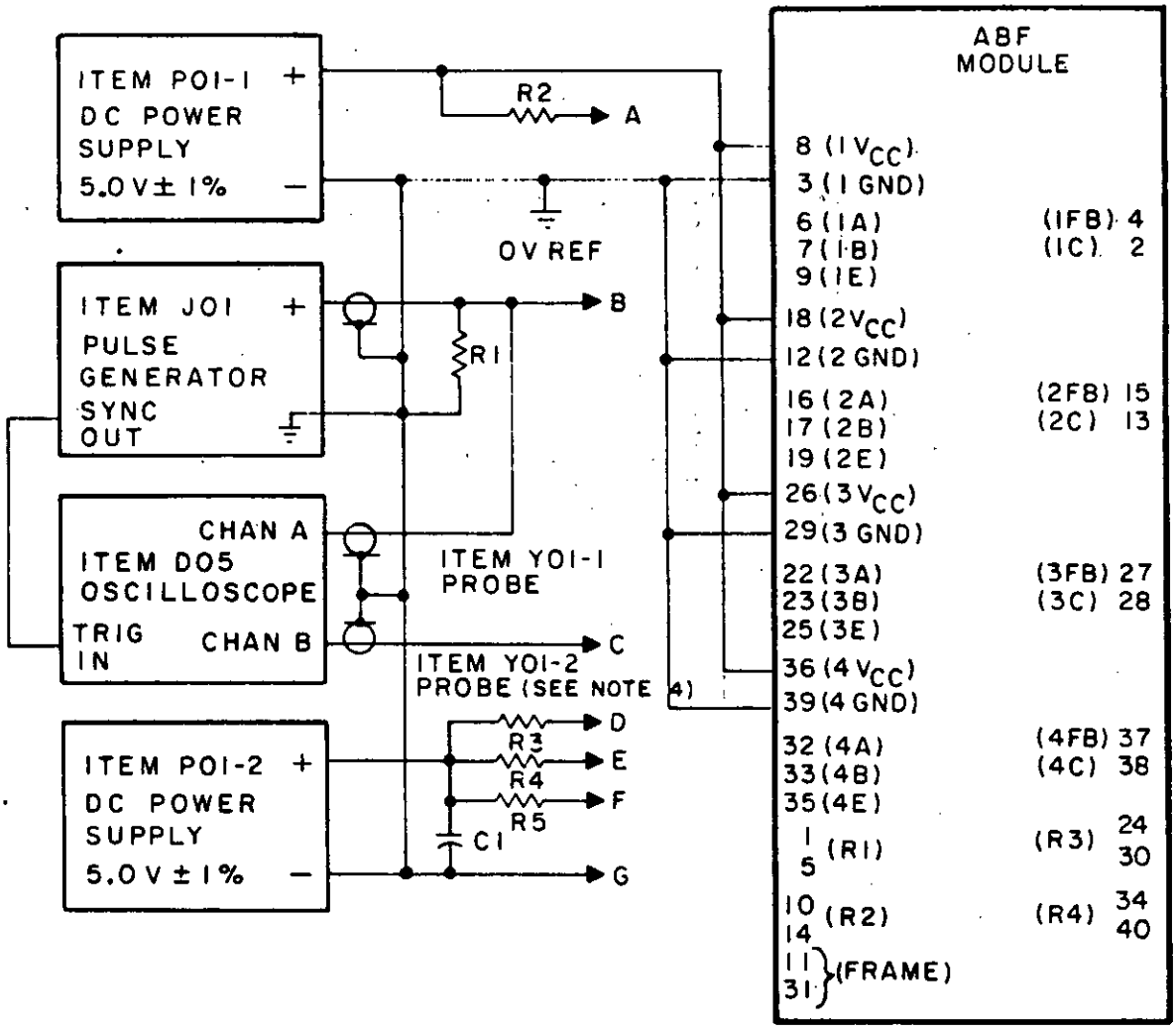
FIGURE 14. Isolation and input capacitance test circuit for tables XII and XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

TABLE XIV. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect							Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
N1	1C Output crosstalk, high level	2	16,22, 32	2	13	28	38	Open	0	550	mV	15,16
N2	2C	13	6,22, 32	13	2	28	38					
N3	3C	28	6,16, 32	28	2	13	38					
N4	4C	38	6,16, 22	38	2	13	28					
N5	1C Output crosstalk, low level	2	16,22, 32	2	13	28	38	6		400		
N6	2C	13	6,22, 32	13	2	28	38	16				
N7	3C	28	6,16, 32	28	2	13	38	22				
N8	4C	38	6,16, 22	38	2	13	28	32				

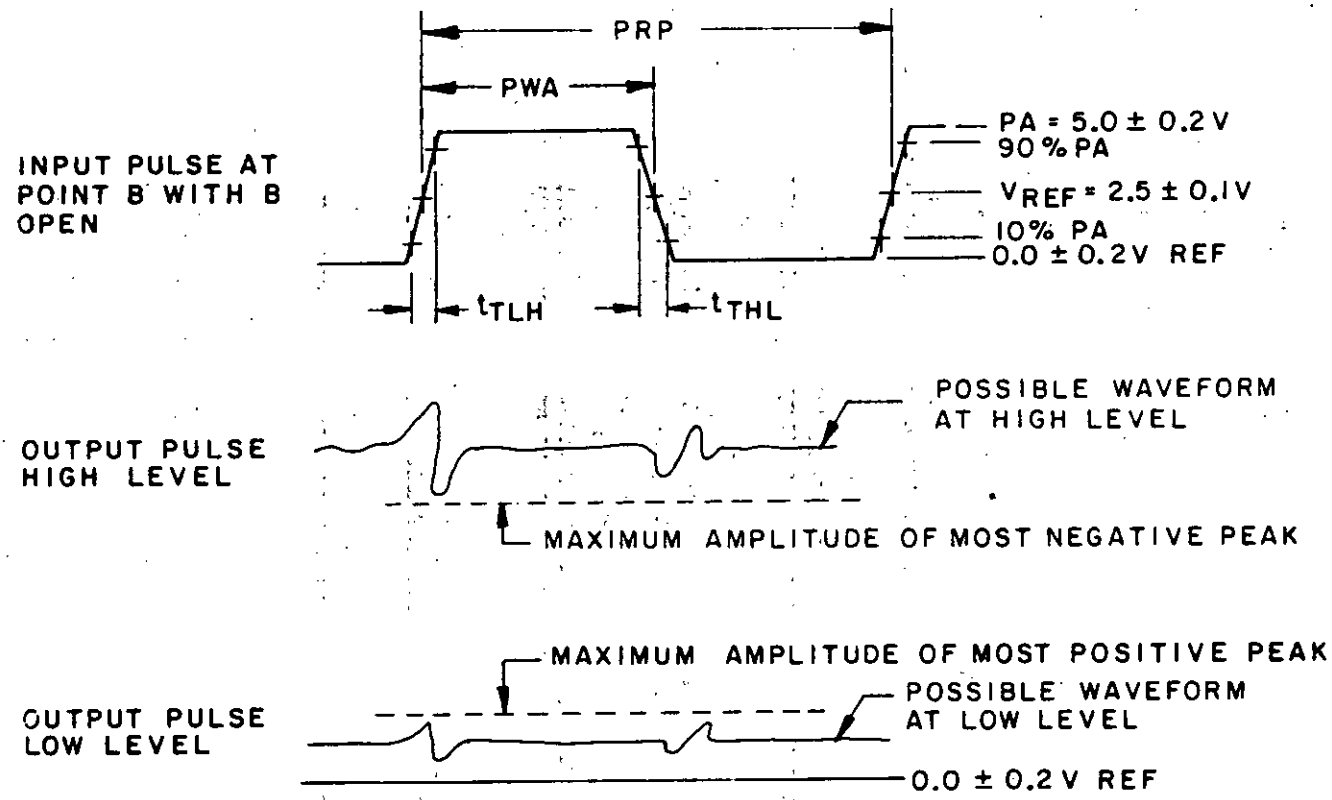
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50 Ω ±5%, 1 W.
2. R2-R5 = 19.1 Ω ±1%, 2 W.
3. C1 = Capacitance should be added to reduce voltage spikes to less than 200 mV.
4. Channel B probe (Item Y01) with AC coupler.

FIGURE 15. Output crosstalk (high-level and low-level) test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:

PRP = 20 ±1 µs.
PWA = 10 ±0.5 µs.
$t_{TLH} = t_{THL}$ = 25 ±5 ns.

FIGURE 16. Output crosstalk (high-level and low-level) test circuit waveforms for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. *Attach any pertinent data which may be of use in improving this document.* If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-28787/152A(EC) Modules, Standard Electronic Digital, Key Code AFB Driver, Lamp or Relay,

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:18Z
Check the source to verify that this is the current version before use.